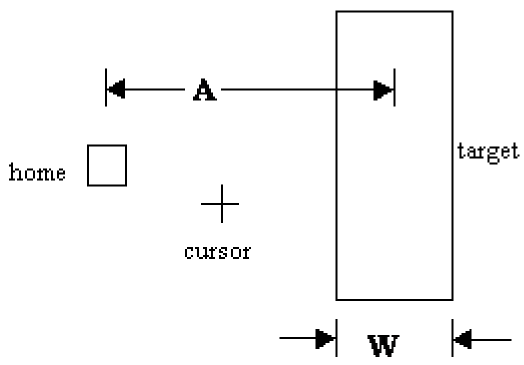
Fitts' Law (Fitts, 1954)

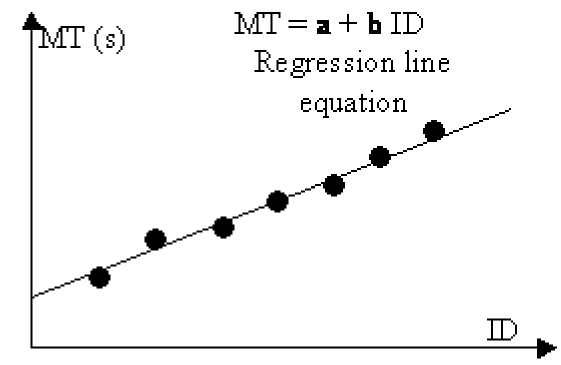
- Fitts' Law predicts that the time to point at an object using a device is a function of the distance from the target object & the object's size.
- The further away and the smaller the object, the longer the time to locate it and point to it.
- Fitts' Law is useful for evaluating systems for which the time to locate an object is important, e.g., a cell and smart phones, a handheld and mobile devices.

Fitts' Law (MacKenzie Formulation)



$$MT = a + b \log_2\left(\frac{A}{W} + 1\right)$$

$$ID = \log_2\left(\frac{A}{W} + 1\right)$$



MT: Movement Time – Time it takes to move cursor from home to target

ID: Index of Difficulty – Metric of tasks difficulty expressed as a measure of human performance in bits per section.

b: measured constant – approximately 200ms/bit.